Approval gates and idempotency

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

Prepare the purpose, prerequisites and materials

> Prepare Approval gates and idempotency: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Explain the **mechanism** and terminology of **approval gate**s and idempotency

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Planning and action selection

Without opening the old lesson, explain one key idea from Planning and action selection. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of planning and action selection

Reference example, not the only acceptable wording: Teacher explanation: Planning proposes steps; execution must still validate permissions and prerequisites for each step. Replanning cannot invent authority.

For read-only reporting, allowed plan is validate -> retrieve -> verify -> answer. If retrieval lacks **evidence**, route to review. Reject write or send steps before they run.

Logs traces and privacy

Without opening the old lesson, explain one key idea from Logs traces and privacy. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of logs traces and privacy

Reference example, not the only acceptable wording: Teacher explanation: Logs support diagnosis, but excessive recording can expose private content. Trace IDs link stages without storing complete sensitive **input**s.

For this policy log trace ID, stage, status, duration and error category; do not log passcodes, student text or teacher answers. Restrict access and declare retention.

Integration tests and fixtures

Without opening the old lesson, explain one key idea from Integration tests and fixtures. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of integration tests and fixtures

Reference example, not the only acceptable wording: Teacher explanation: Integration tests exercise connected components and failure propagation. Fixtures supply controlled **input** records and mocked service outcomes.

Trace request validation, session permission, lookup and **output** validation together. Include absent sources and failing dependencies; verify that denied requests never invoke restricted lookup.

Two 45-minute sessions. **Approval gate**s bind side effects to an actor, exact version and destination. Idempotency prevents repeated requests with the same key from repeating a committed effect.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **approval gate**s and idempotency

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

Prerequisite answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

End goal for the pictorial case

Explain **approval gate**s and idempotency using the deciding step in this fictional service.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain and model the mechanism

> Model Approval gates and idempotency using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Approval gates bind side effects to an actor, exact version and destination. Idempotency prevents repeated requests with the same key from repeating a committed effect.

Explain Approval gates and idempotency · 1

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

The mock submission requires approval for request version v1 and destination D. Store key -> committed result. Reusing the same key with the same payload returns that result; the same key with different payload is rejected.

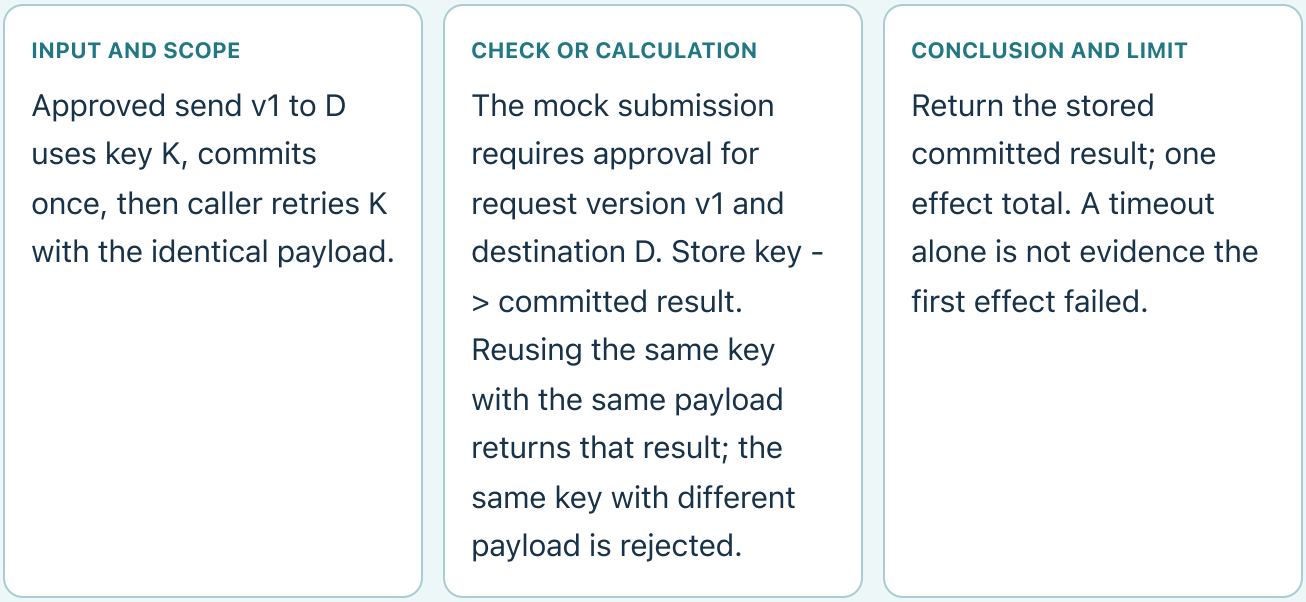
Explain Approval gates and idempotency · 2

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Idempotency key	An identifier for deduplicating one logical operation
Approval gate	A required permission check before a side effect
Payload conflict	Different operation content under the same key

Approval gates and idempotency — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Explain Approval gates and idempotency · 3

Guided: Return the stored committed result; one effect total. A timeout alone is not **evidence** the first effect failed.

Qualification: A key prevents duplicate application effects only within the implemented persistence and downstream guarantees; it is not universal exactly-once delivery.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Approved send v1 to D uses key K, commits once, then caller retries K with the identical payload.

Return the stored committed result; one effect total. A timeout alone is not **evidence** the first effect failed.

Model the reasoning aloud · 4

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Model the reasoning aloud · 5

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
from copy import deepcopy

class MockLedger:
    """In-memory classroom deduplication, not durable exactly-once delivery."""

    def __init__(self):
        self.committed = {}
        self.effects = []

    def submit(self, key, payload, approval):
        if approval != payload:
            raise PermissionError('exact approval required')
        if key in self.committed:
            (original, result) = self.committed[key]
            if original != payload:
                raise ValueError('idempotency conflict')
            return deepcopy(result)
        result = {'operation_id': key, 'status': 'committed'}
        self.effects.append(deepcopy(payload))
        self.committed[key] = (deepcopy(payload), deepcopy(result))
        return deepcopy(result)

ledger = MockLedger()
payload = {"version": "v1", "destination": "D"}
ledger.submit("K", payload, payload)
ledger.submit("K", payload, payload)
print(len(ledger.effects))
```

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Approval gates and idempotency

One approval, one reservation. A desk tool needs approval for the exact request and deduplicates repeated request keys.

The facts we will use

Supplied facts

K9 reserves desk 4 and is approved once. Two copies of K9 arrive.

AI ACADEMY / AI-6/15

Approval gates and idempotency

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check approval

The approved identifier and desk match K9.



2. Execute first copy

No result is stored yet, so perform the approved reservation once.



3. Handle duplicate

Return the stored K9 result without another reservation.

CHANGE ONE THING

A new K10 asks for desk 5 without approval.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check approval	The approved identifier and desk match K9.
2. Execute first copy	No result is stored yet, so perform the approved reservation once.
3. Handle duplicate	Return the stored K9 result without another reservation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Approval answers whether an action may proceed. Idempotency prevents repeated handling of that action from creating duplicate effects.

What this example does not establish

Keep the stated limit

Deduplication requires reliable stored keys; this illustration does not prove crash-safe exactly-once execution.

Change one thing in the pictured case

Change one thing

Change: A new K10 asks for desk 5 without approval.

Predicted result

Prediction: Stop K10 at the **approval gate**.

Why the result follows

Reason: K9's approval does not apply to a different action.

Ask while showing the picture

Ask while showing the picture · 6

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Approval gates and idempotency, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Reduce help deliberately

Completion practice

Completion task: Approved send v1 to D uses key K, commits once, then caller retries K with the identical payload. Show the intermediate steps and give the conclusion with its scope.

Reasoning to listen for

Teacher check: Teacher explanation: Return the stored committed result; one effect total. A timeout alone is not **evidence** the first effect failed.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Approval gate**s and idempotency. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: A key prevents duplicate application effects only within the implemented persistence and downstream guarantees; it is not universal exactly-once delivery.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Trace the two deliveries of approved K9. How many reservations should occur, and why can a new K10 for desk 5 not borrow K9’s approval?

Reasoned teacher answer

The approved identifier and desk match K9. No result is stored yet, so perform the approved reservation once. Return the stored K9 result without another reservation. Changed case: Stop K10 at the **approval gate**.

K9′s approval does not apply to a different action.

Changed-case reasoning: Stop K10 at the **approval gate**. K9′s approval does not apply to a different action.

Collect each learner′s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the investigation and diagnose misconceptions

> Run the Approval gates and idempotency investigation and use the first unsupported step to choose a teaching response.

The worked result establishes every future case: A key prevents duplicate application effects only within the implemented persistence and downstream guarantees; it is not universal exactly-once delivery.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Approval gates and idempotency questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Guided: Return the stored committed result; one effect total. A timeout alone is not **evidence** the first effect failed.

Qualification: A key prevents duplicate application effects only within the implemented persistence and downstream guarantees; it is not universal exactly-once delivery.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

Workbook question · W1

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher reasoning · W1

Teacher explanation: **Approval gate**s bind side effects to an actor, exact version and destination. Idempotency prevents repeated requests with the same key from repeating a committed effect.

The mock submission requires approval for request version v1 and destination D. Store key -> committed result. Reusing the same key with the same payload returns that result; the same key with different payload is rejected.

W2 Trace the worked case

Workbook question · W2

Approved send v1 to D uses key K, commits once, then caller retries K with the identical payload. Show the intermediate steps and give the conclusion with its scope.

Teacher reasoning · W2

Teacher explanation: Return the stored committed result; one effect total. A timeout alone is not **evidence** the first effect failed.

W3 Audit the limit

Workbook question · W3

A report omits this qualification: A key prevents duplicate application effects only within the implemented persistence and downstream guarantees; it is not universal exactly-once delivery.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher reasoning · W3

Teacher explanation: A key prevents duplicate application effects only within the implemented persistence and downstream guarantees; it is not universal exactly-once delivery.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W4 Apply a new case independently

Workbook question · W4

Key J already committed room A. A retry uses key J but room B. What should happen?

Teacher reasoning · W4

Teacher explanation: Reject key/**payload conflict**; do not silently send B or return A as though it represented B. Obtain version-correct approval for a new operation.

W5 Design a revealing follow-up

Workbook question · W5

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher reasoning · W5

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A key prevents duplicate application effects only within the implemented persistence and downstream guarantees; it is not universal exactly-once delivery.

The new case should test this particular **mechanism** rather than merely restate the original result.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4: Reject key/**payload conflict**; do not silently send B or return A as though it represented B. Obtain version-correct approval for a new operation. Require intermediate reasoning and the stated scope.

Fresh retry: Key M committed recipient X; retry changes recipient Y. Expected reasoning: Reject mismatch and preserve original result. Idempotency does not authorise a new recipient.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Independent question · withhold the key

Selected case: Key J already committed room A. A retry uses key J but room B. What should happen?

Selected case answer

Teacher key · Selected case answer

Teacher explanation: Reject key/**payload conflict**; do not silently send B or return A as though it represented B. Obtain version-correct approval for a new operation.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

TEACHER GUIDE · 6 OF 9

Connect the project, support and next lesson

> Connect Approval gates and idempotency to Grounded Learning Service, adapt support and plan a later changed-case check.

Add a labelled **approval gate**s and idempotency **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **approval gate**s and idempotency.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Timeouts retries and cancellation. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-6/16, ai-6/18, ai-6/21. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Approval gates and idempotency lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Explain the **mechanism** and terminology of **approval gate**s and idempotency

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Approved send v1 to D uses key K, commits once, then caller retries K with the identical payload. Show the intermediate steps and give the conclusion with its scope.

Supported practice answer

Reasoned practice answer: Teacher explanation: Return the stored committed result; one effect total. A timeout alone is not **evidence** the first effect failed.

Check every learner before discussion: Use the worked case for **Approval gate**s and idempotency. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: A key prevents duplicate application effects only within the implemented persistence and downstream guarantees; it is not universal exactly-once delivery.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: A new K10 asks for desk 5 without approval. Require a **prediction** and a reason before comparing. Expected result: Stop K10 at the **approval gate**. Reason: K9’s approval does not apply to a different action.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the

new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

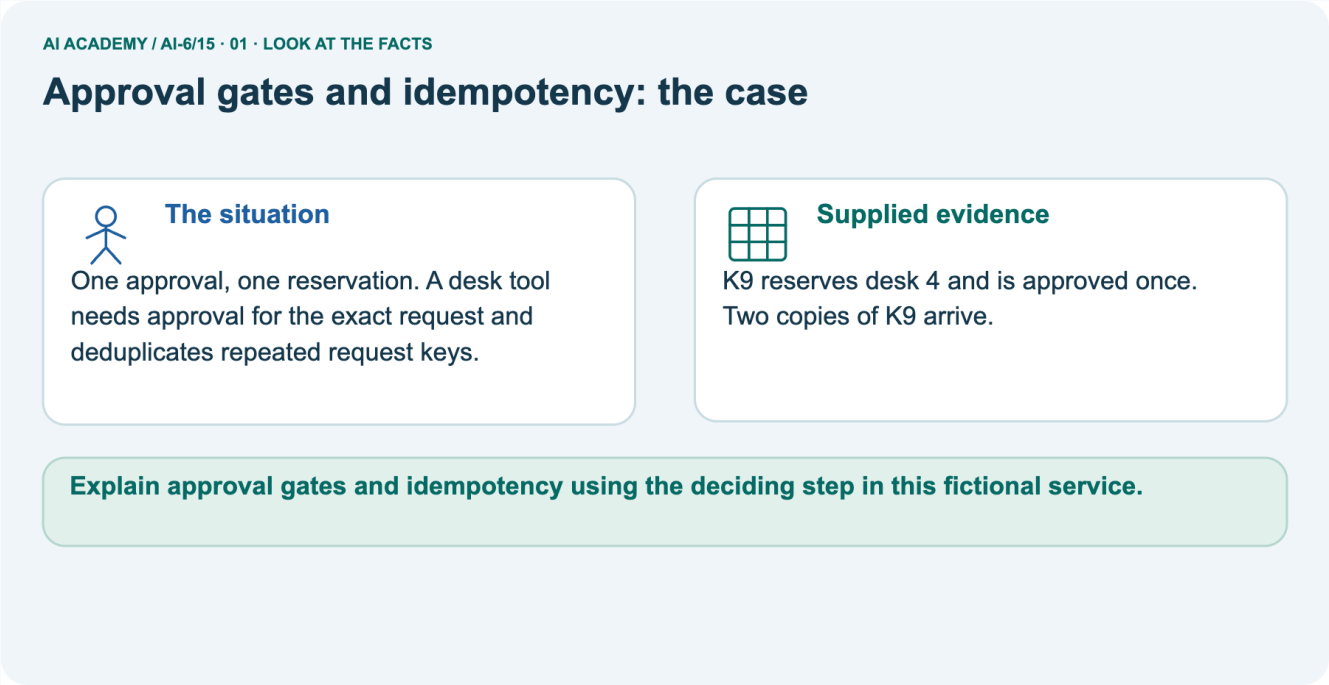
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Approval gates and idempotency graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Approval gates and idempotency: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

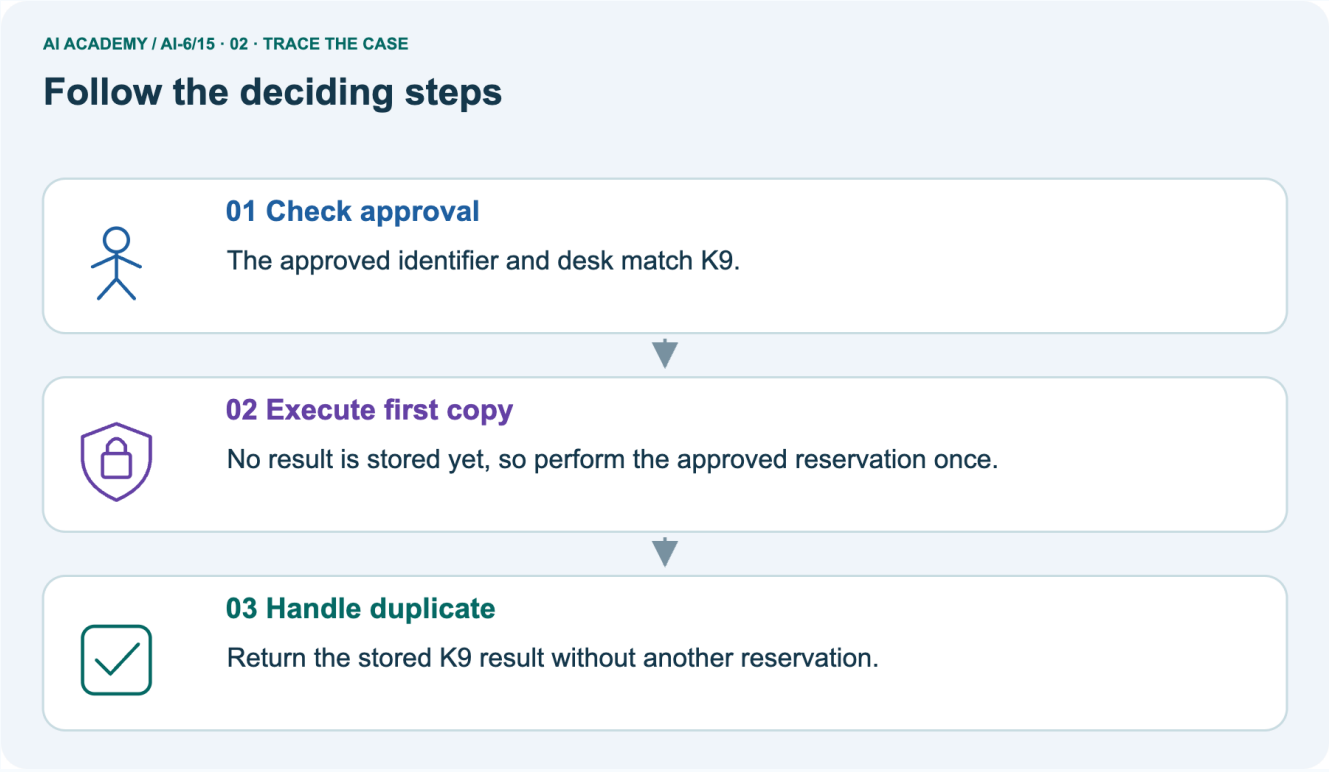
One approval, one reservation. A desk tool needs approval for the exact request and deduplicates repeated request keys. Supplied facts: K9 reserves desk 4 and is approved once. Two copies of K9 arrive.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

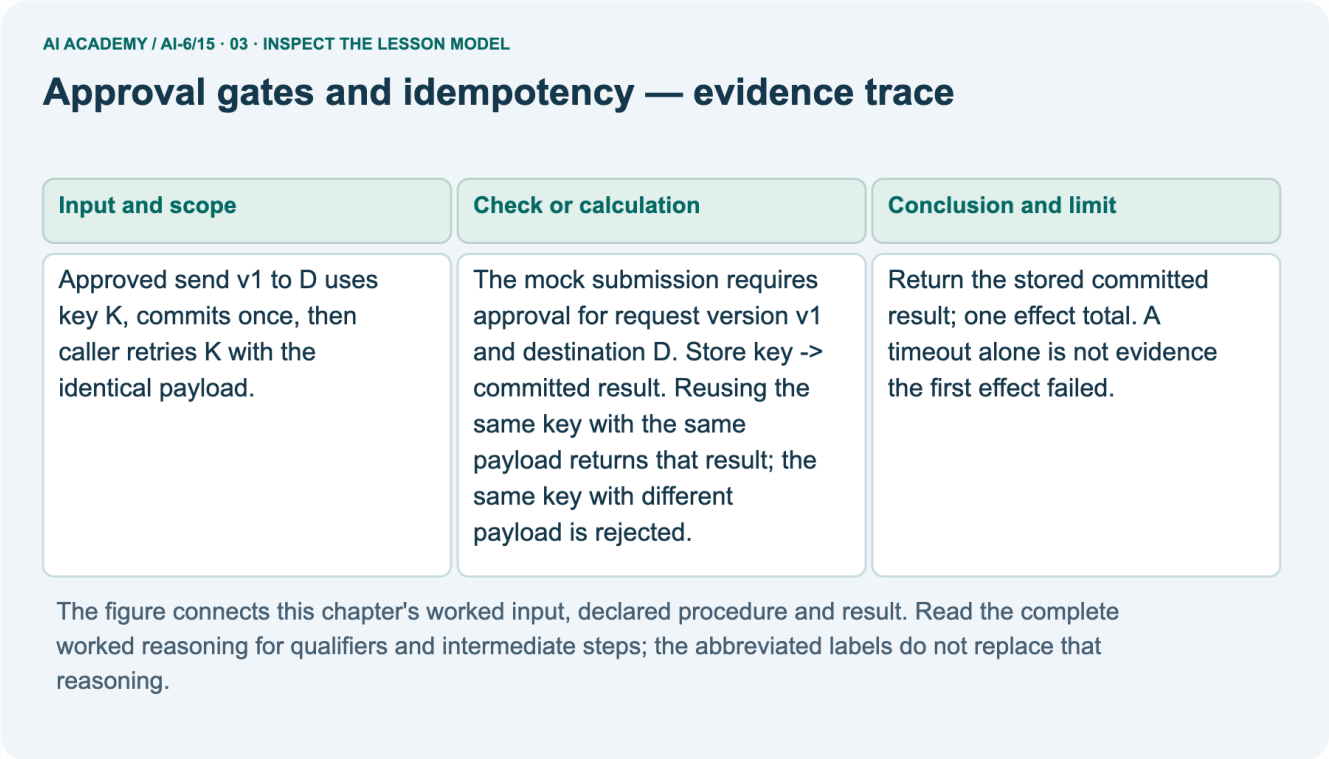
Check approval: The approved identifier and desk match K9. → Execute first copy: No result is stored yet, so perform the approved reservation once. → Handle duplicate: Return the stored K9 result without another reservation.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Approval gates and idempotency — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Approval gates and idempotency — **evidence** trace.

Approved send v1 to D uses key K, commits once, then caller retries K with the identical payload. | The mock submission requires approval for request version v1 and destination D. Store key -> committed result.

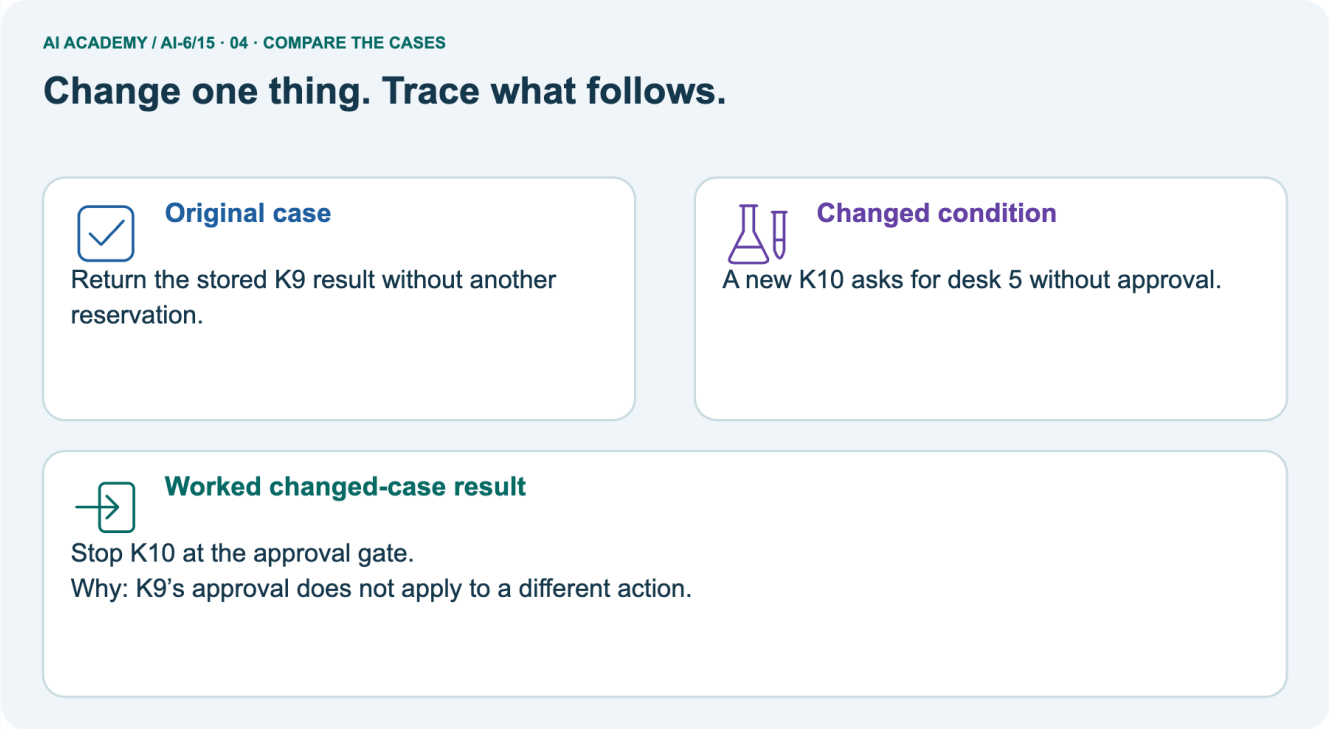
Reusing the same key with the same payload returns that result; the same key with different payload is rejected. | Return the stored committed result; one effect total. A timeout alone is not **evidence** the first effect failed.

> Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Return the stored K9 result without another reservation. Changed condition: A new K10 asks for desk 5 without approval. Expected result: Stop K10 at the **approval gate**.

Reason: K9's approval does not apply to a different action.

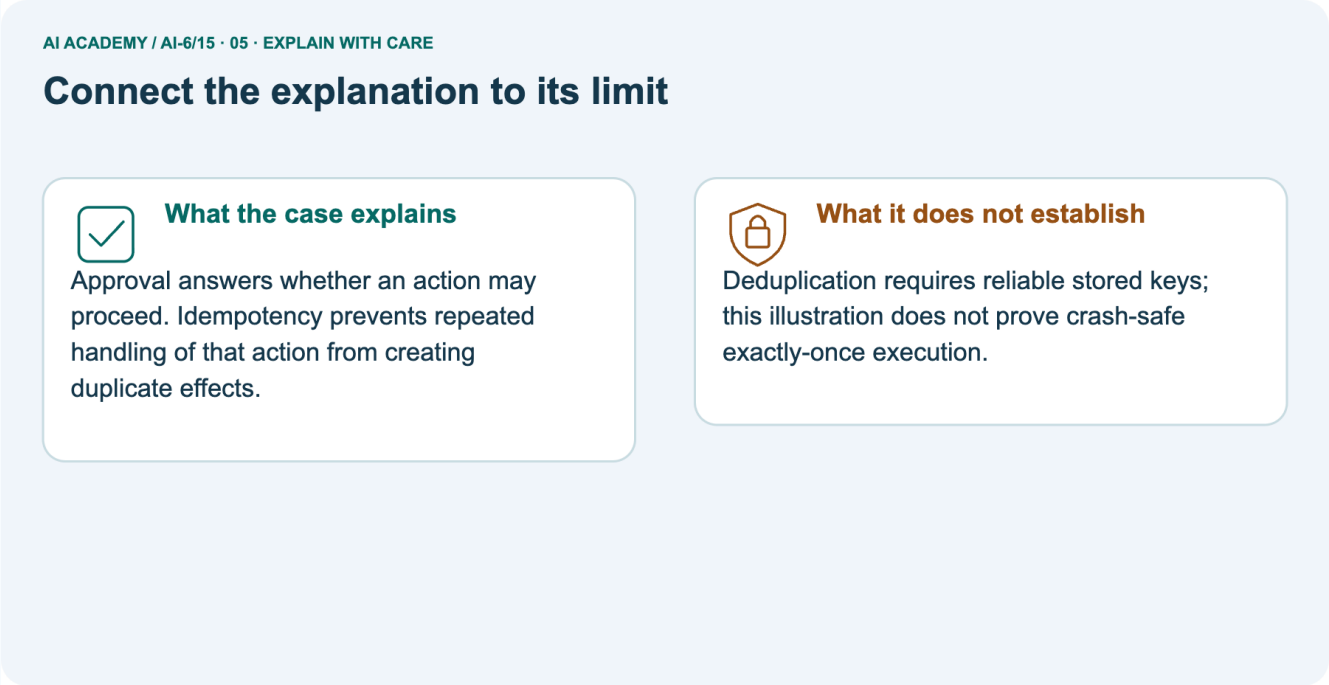
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: K9’s approval does not apply to a different action.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Approval answers whether an action may proceed. Idempotency prevents repeated handling of that action from creating duplicate effects.

Limit: Deduplication requires reliable stored keys; this illustration does not prove crash-safe exactly-once execution.

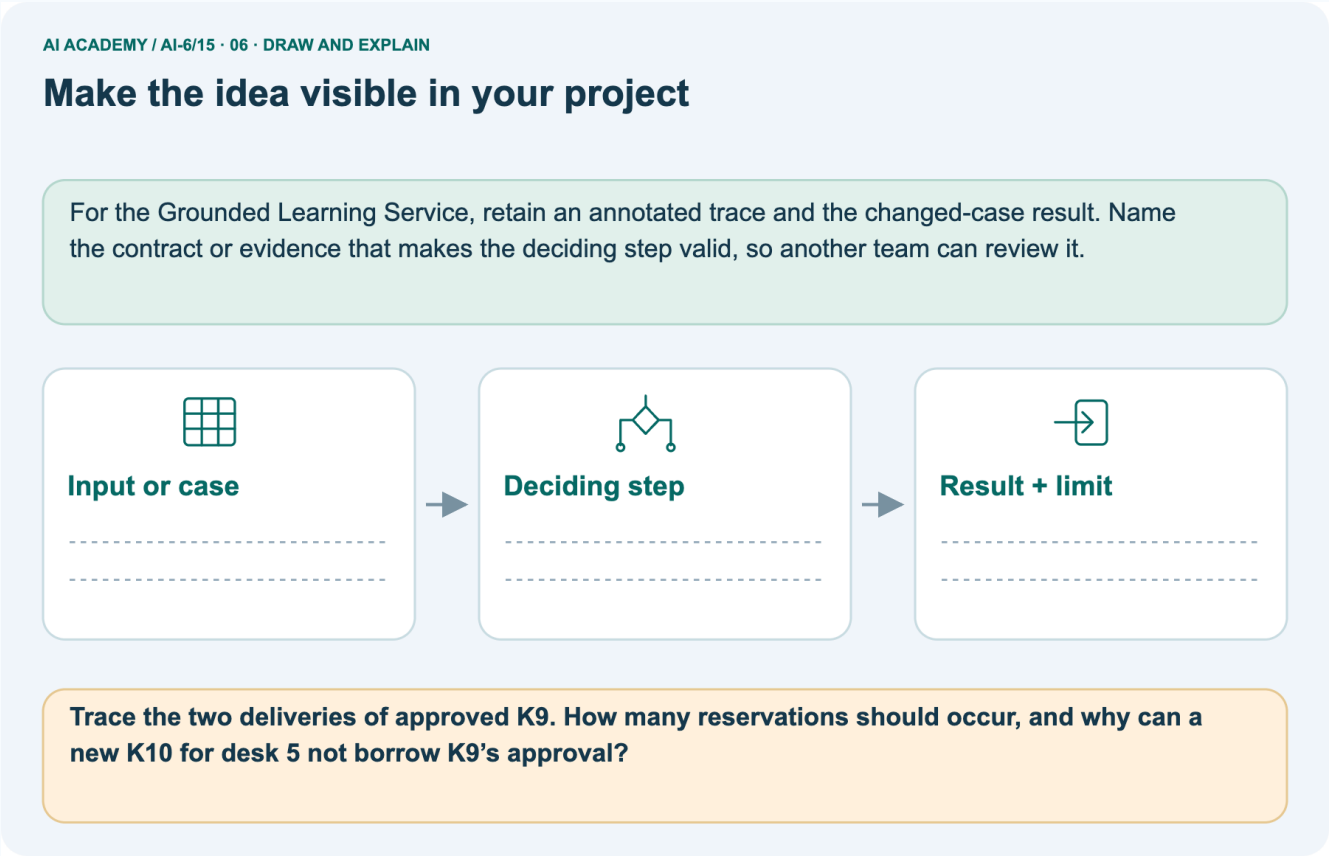
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Deduplication requires reliable stored keys; this illustration does not prove crash-safe exactly-once execution.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Trace the two deliveries of approved K9. How many reservations should occur, and why can a new K10 for desk 5 not borrow K9’s approval?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The approved identifier and desk match K9. No result is stored yet, so perform the approved reservation once. Return the stored K9 result without another reservation.

Changed case: Stop K10 at the **approval gate**. K9’s approval does not apply to a different action.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.